Cuong Q. Pham

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PROFESSIONAL INTERESTS

I have a strong interest in data-driven healthcare research integrating signal processing and machine learning. **Current Research Focus**: Digital phenotyping, AI for mental health, multimodal biosignal analysis, biomedical signal processing, and development of Software as a Medical Device (SaMD).

EDUCATION

University of Illinois Urbana-Champaign (UIUC) — United States
M.S. in Electrical and Computer Engineering — *now*

Ritsumeikan University — Japan
M.E. in Information Science and Engineering — *2023*

VNU-HCM University of Technology — Vietnam
B.E. in Physics Engineering — *2020*

WORK EXPERIENCE

Department of Electrical and Computer Engineering, UIUC — IL, USA
Graduate Student — *Aug 2026 – now*

- **Project**: *"LittleBeats"*. **Funding(s):** NIH. **My Role:** Research Assistant
  - **Collaboration**: **PI(s):** Prof. Nancy McElwain (ILSD, UIUC), Prof. Mark Hasegawa-Johnson (ECE, UIUC), Prof. Bashima Islam (ECE, WPI)
  - **Responsibilities**: Analyzed biosignal data collected from infants and developed machine learning models to characterize infant physiological patterns.

VNU HCMIU x VinUniversity — Vietnam
Tech-lead — *Oct 2025 – Jul 2026*

- **Project**: *"Neuroplasticity Tracking During Intervention For Depression and Anxiety: A Longitudinal, Cognitive-guided Digital Phenotyping Study"*. **Funding:** IBRO/Wellcome. **My Role:** Lead Proposal Writing & Project Execution.
  - **Collaboration:** VinUniversity, HCMIU, RMIT VN, Nguyen Tri Phuong Hospital, Phuong Nam Hospital.
  - **Responsibilities**: Led the development of a mobile digital CBT platform and conducted a longitudinal study (n=60, 4.5 months) on Vietnamese mental health patients to investigate neuroplasticity changes. [Link]

School of Computer Science, University of Birmingham — Birmingham, UK
Visiting Research Student (Mentor: Dr. Melanie Jouaiti) — *Jul – Sep 2025*

- **Project:** *"Bayesian Commorbidity-aware Modeling For Interpretable Speech Features Based Depression Severity Detection"*. **Funding:** IEEE SPS. **My Role:** Lead Proposal Writing & Project Execution.
  - **Responsibilities**: Developed and evaluated an interpretable, comorbidity-aware disentangled learning framework to model links between speech features and depression severity.

VinUni-Illinois Smart Health Center, VinUniversity — Hanoi, Vietnam
Graduate Student (Supervisor: Dr. Hieu Pham) — *Aug 2024 – Jul 2026*

- **Project**:*"Towards Improving Longitudinal Digital Phenotyping Representation Framework for Wellbeing Modeling"*. **Funder:** VISHC. **Duration:** Dec 2024 - Jul 2025. **My Role:** Project Execution
  - **Responsibilities**: Developed and evaluated a novel framework for unified digital phenotype data formulation to improve various longitudinal well-being modeling tasks.

- **Project**: *"Developing and Implementing Digital Mental Health Assessment Tools in Low-Resource Settings".* **Funder:** VISHC. **Duration:** Apr - Dec 2025. **My Role:** Tech-lead, Project Coordinator
  - **Collaboration:** VinUniversity, VNU HCMIU, RMIT Vietnam, Nguyen Tri Phuong Hospital, Menthy Clinics.
  - **Responsibilities**: Developed a mobile application for multimodal digital phenotype acquisition, integrating physiological signals, behavioral indicators, and psychological self-reports; conducted short-term data collection with Vietnamese mental health patients (n=100, 15 days).

School of Biomedical Engineering, VNU HCMIU HCMC, Vietnam

Research Assistant (Advisor: Dr. Huong Ha) *Nov 2023 – Jul 2024*

- **Project**: *"Investigation on establishing a large database of EEG and video recordings of Vietnamese people in application for intelligent control and primary motor rehabilitation in epileptic patients".* **Funding:** Vietnam Ministry of Science and Technology (KC-4.0-07/19-25). **My Role:** Tech Lead @ HCMIU Team
  - **Collaboration**: VNU HCMIU, VNU UET, EMOTIV, 175 Military Hospital.
  - **Responsibilities:** (1) tailored Motor Imagery experiment and processing pipelines to Vietnamese subjects, benchmark predictive modelings, deployed web apps for neuro-feedback; (2) developed app for data acquisition and real-time mouse control system, integrating into into cross-regional project platform.

HATO Medical Technologies ApS Odense, Denmark

Biosignal Researcher *Jun 2022 – Nov 2023*

- Job responsibilities
  - **Cardiology Research**: (1) worked closely with cardiologists to establish standardized CVDs labeling outcomes tailored to specific use cases at a local Danish ED; (2) conduct literature reviews for evidence-based decision making on cardiology practice, prepare technical materials and grant proposals/fundings.
  - **Data Pipeline**: (1) collect, preprocess clinical data from public and closed sources; (2) worked closely with software developers to integrate serialization pipeline into the backend architecture of the in-house product.
  - **AI/ML Development**: (1) implement a Cloud-based internal data management system with interactive web app and tested its streamline workflow; (2) evaluate and deployed time-series predictive modeling for abnormalities detection and interpretation; inspected and ensured the solution meet technical requirements.

Biological Engineering Laboratory, Ritsumeikan University Shiga, Japan

Graudate Research Assistant (Advisor: Dr. Kashihara Koji) *Oct 2021 – Aug 2023*

- **Project**: *"A Hybrid Controller for Multiple Drug Infusion in Heart Failure".* **Duration:** Sep 2021 - Mar 2022
  - **Responsibilities:** (1) develop a hybrid controller to regulate CO and MAP within during drug infusion using ML model with drug inputs; (2) evaluate on a mathematical modeling responses of dogs with heart-failure dataset.
- **Project**: *"Remote Photoplethymosgraph Assessment".* **Duration:** Apr 2022 - Aug 2023
  - **Responsibilities:** (1) design framework to extract high-quality forehead rPPG signal via autoconder networks; (2) conduct rPPG dataset on Japanese students and evaluate HR on different camera configurations; (3) design framework to assess stiffness-based waveform feature via deep 3D CNN.

GTOPIA Vietnam. Ltd HCMC, Vietnam

Signal Processing Engineer *Jan – Jun 2020*

- Job responsibilities
  - **Wearable Research:** (1) design pipeline for raw data aggregation and hemodynamic signal processing for in-house wearable product; (2) conduct experiments on wristbands's performance under different usage scenarios.
  - **EHR Data Collection:** (1) collaborate with Ho-Chi-Minh-Heart-Institute for large-scale clinical data acquisition; (2) categorize and digitalize health records of administered patients with cardiovascular diseases

Biomedical Electronics Laboratory, Shibaura Institute of Technology Tokyo, Japan
Research Intern (Advisor: Dr. Shinichiro Kanoh) *Sep – Nov 2019*

- **Project**: *"Design Motor Imagery Protocol To Adapt ERD/ERS Response."*
  - **Responsibilities**: conduct experimental analysis on EEG visualization for motor cortex response and how to conduct neuro-feedback during BCI experiment.

Faculty of Applied Science, VNU-HCM University of Technology HCMC, Vietnam
Undergraduate Research *Jun 2018 – Mar 2019*

- **Project**: *"Exercise Physiology Application: Cardiac Endurance Training for Students by Stationary Bike".* **Collaboration**: HCMC Institute of Biomedical Physics. **Funder**: FAS, HCMUT. **My Role:** Technical Lead.
  - **Responsibilities**: (1) to design circuits for workload adjustment adapting to the biker's heart rate; (2) to evaluate VO2max improvement on healthy college students (n=15, 2 months) over endurance training course.

FUNDINGS

[Oct 2025 - Jun 2026] Neuroplasticity Tracking During Intervention For Depression and Anxiety: A Longitudinal, Cognitive-guided Digital Phenotyping Study

- Funder: Neuroscience Capacity Accelerator for Mental Health 2025 (IBRO/Wellcome)
- Amount: 60,000 USD
- Role: Lead Proposal Writing, Project Tech-Lead.

[Jan - Sep 2025] Multimodal Data Modeling Framework for Early Depression Detection

- Funder: Signal Processing Mentorship Academy 2025 (IEEE SPS)
- Amount: 3,500 USD
- Role: Lead Proposal Writing & Project Execution.

TEACHING

Teaching Assistant @ CECS, VinUniversity

- COMP1010 Introduction to Programming (Fall'25)
- COMP1020 Object-Oriented Programming and Data Structures (Spring'25, Spring'26)
- COMP3040 Computer Vision (Fall'24, Fall'25)

Teaching Assistant @ CISE, Ritsumeikan University

- Experiments in Artificial and Natural Intelligence (Fall'22)

AWARDS

- **[Jul 2026]** Vingroup Science and Technology Scholarship Program for Overseas Study; by Vingroup.
- **[Mar 2026]** Top 30 QVIC 2026; by Qualcomm Vietnam.
- **[Feb 2026]** Travel Grants (600 USD); by IBRO.
- **[Jan 2025]** Travel Grants (3500 USD); by IEEE SPS.
- **[Aug 2024]** Fully-funded Fellowship; by VISHC, VinUniversity.
- **[Aug 2022]** 2nd prize in Kyoto Startup Weekend Competition; by Techstars.
- **[Mar 2022]** GAKKAI scholarship; by Ritsumeikan University.
- **[Sep 2021]** Fully-funded MEXT Scholarship; by Japanese Government.

PATENTS

- The brain-computer interface system controls the mouse cursor based on motor imagination and neural states. *Intellectual Property Office of Vietnam* (Under Review)

PUBLICATIONS & MANUSCRIPTS

Manuscript Under Review

- A multimodal digital phenotyping dataset for depression and anxiety assessment under free-living conditions *Scientific Data. (Under Review)*
- Cue Design and Semantic Feedback Improve Temporal Consistency of Motor Imagery in EEG-Based Brain–Computer Interfaces. *Frontiers (Under Review)*

2026

- **Cuong Q. Pham** *et al.* A multimodal digital phenotyping dataset for depression and anxiety assessment under free-living conditions (1.0.2) [Data set]. *Zenodo. DOI: 10.5281/zenodo.21969065* [dataset] [github]
- **Pham, Q. C.**, Vu, H. T. D., Ha, T. T. H., & Pham, H. H. (2026). Neurai-VN Benchmark: Standardized Machine Learning Models for Multimodal Digital Phenotyping in Mental Health Classification. *arXiv preprint arXiv:2607.25232.* [arxiv] [github]
- Vu, H. T. D., **Pham, Q. C.**, & Pham, H. H. (2026, July). VISHC at PsyDefDetect: Mitigating Data Scarcity in Psychological Defense Classification with Context-Aware Synthetic Augmentation. *In Proceedings of the BioNLP 2026 (Shared Tasks) (pp. 77-86)* [paper] [github]

TALKS

- [**Apr. 2026**] Neuroplastic Change During Cognitive-Behavioral Therapy for Depression and Anxiety: A Longitudinal Study @ *IBRO Complexity in Mental Health Workshop (Sicily, Italy).*
- [**Dec. 2024**] Development and Evaluation of Multimodal AI Framework for Mental Health Assessment: A Preliminary Study @ *MCABBD workshop, Brain Informatics 2024 (Bangkok, Thailand)*
- [**Jun. 2024**] Evaluation of Cue-based Protocol Implementations in Motor Imagery - based Brain-Computer Interface Experiments @ *NeuroCoB/Brainconnects 2024 (Putrajaya, Malaysia).*

ACADEMIC ACTIVITIES

School Projects

- [**Fall 2022**] WasteWise @ GSISE, Ritsumeikan University
  - Team of 6 collaborate with TH Nürnberg (Germany); develop an AI-based mobile app for trash-bins time collection recommendation in public spaces using crowdsourcing dataset.
  - **Achievement**: Deployed app and evaluated on the pilot data in Shiga and Kyoto city.
- [**Summer 2022**] Pic2Fit @ KYOTO Design Lab, Kyoto Institute of Technology
  - Designed a proof-of-concept virtual clothes fitting application tailored for small shops in Kyoto, Japan.
  - **Achievement**: Awarded at Kyoto Startup Weekend Competition.
- [**Fall 2018**] Stationary Bike @ VNU-HCM University of Technology
  - Designed circuits for workload adjustment adapting to the biker's heart rate; collaborated with HCMC Institute of Biomedical Physics to evaluated VO2max improvement on students over endurance training course.
  - **Achievement**: The system is integrated into laboratory experiment course for students in afterwards cohorts.
- [**Spring 2017**] Pet Feeder
  - Tech-lead freelance team to design the low-cost automated pet-feeding system; conducted mechanical design and material 3D-printing, developed electrical circuits and platform for IoT user control.
  - **Achievement**: Delivered MVP to the reserved customers.

Communitiy Involvement

- [**Dec 2024**] Conference Staff @ ACML 2024 (Hanoi, Vietnam).
- [**Jan '203**] Teaching Staff @ Ritsumeikan Junior High (Kyoto, Japan).
- [**Oct 2022**] Conference Staff @ IEEE/RSJ IROS 2022 (Kyoto, Japan).

SELECTED SKILLS

- **Programming**: Python, JavaScript, TypeScript, MATLAB, R, SQL, C#, Bash/Linux
- **Machine Learning**: OpenCV, Scikit-learn, LightGBM, XGBoost, Keras, PyTorch, Lightning
- **System Development**: Web (React, CSS), Mobile (React Native), Backend (Flask, FastAPI)
- **Cloud**: AWS (S3, EC2, Lambda), GCP
- **Databases**: MySQL, PostgreSQL, MongoDB, Firebase
- **Tools**: Git, Docker, Jira, Streamlit, Lab Streaming Layer
- **Miscellaneous**: Data Analysis (scipy, pandas, ggplot2, dplyr), Bio-Signal Experimentation (ECG, EEG, PPG, EMG, wearable/bio-sensors), Signal Processing (spectral & time-frequency analysis, transformations, filtering), Circuit (Arduino, Raspberry Pi)
- **Language**: Vietnamese (native), English (professional, IELTS 7.0)

REFERENCE

Hieu Pham, Ph.D.
Assistant Professor, College of Engineering & Computer Science (CECS) &
Scientific Director, Entrepreneurship Lab (E-lab),
PI at VinUni-Illinois Smart Health Center, VinUniversity.
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